A Sample Article Using IEEEtran.cls for IEEE Journals and Transactions

IEEE Publication Technology, *Staff, IEEE,*

***Abstract*—This document describes the most common article elements and how to use the IEEEtran class with LATEX to produce files that are suitable for submission to the IEEE. IEEEtran can produce conference, journal, and technical note (correspondence) papers with a suitable choice of class options.**

***Index Terms*—Article submission, IEEE, IEEEtran, journal, LATEX, paper, template, typesetting.**

I. Introduction

WITH climate change, renewable energy has become a crucial part of the energy transition in countries around the world, and the penetration rate of wind power has continued to rise accordingly. However, the randomness and intermittency of wind speeds, as well as forecasting errors, cause frequent fluctuations in the grid-connected power of wind farms [1], which in turn affect power quality, dispatch reliability, and grid security [2], [3]. Regarding power fluctuations caused by wind power, existing control methods fall into two main categories: one involves power smoothing solely by adjusting the wind turbine's pitch angle or axial induction factor. This method often results in a loss of wind energy utilization rate [4] and increases fatigue on the mechanical components of wind turbines [5]. The other category involves introducing energy storage facilities to form a Wind-Storage Integrated System (WSIS) in conjunction with the wind farm. By leveraging the power regulation capabilities of wind turbines and the rapid charging and discharging capabilities of energy storage, WSIS suppresses grid-connected power fluctuations while increasing power generation revenue, and has become a key solution for integrating high proportions of wind power into the grid [6].

WSIS control requires coordinating wind turbines and energy storage systems with different dynamic characteristics and control frequencies: to avoid wind energy losses and mechanical fatigue caused by frequent adjustments to the wind turbines, the turbines typically optimize their overall power output on a slower time scale; meanwhile, the energy storage system tracks a smooth target on a faster time scale to achieve a smooth overall output [7]. Mainstream wind turbine control methods each have their own limitations and shortcomings: PID control relies on linearized model parameters; in wind power environments characterized by strong nonlinearity, high inertia, and time delays, it suffers from poor robustness, and parameter tuning is time-consuming and difficult; Model predictive control and rule-based control rely on accurate wake models [8], wind speed forecasts, and battery models; in large-scale, highly uncertain environments, they suffer from complex modeling, high computational overhead [9], [10], and limited generalization ability [11]. Furthermore, there is a clear conflict between maximizing economic benefits and power fluctuation constraints [12], which may result in overly conservative strategies or frequent violations of safety constraints.

To address the above issues, reinforcement learning (RL)—as a model-free approach that directly learns control policies through interaction with the environment without the need for prior modeling of system dynamics—offers a new approach to controlling complex WSIS systems [7], [13]. Hierarchical Safe Reinforcement Learning (HSRL) utilizes different control time scales to decompose upper-level wind turbine scheduling and lower-level energy storage smoothing into mutually cooperative decision-making processes, while ensuring compliance with control constraints and objectives [14]. However, existing methods still face challenges such as reward boundaries relying on manual experience, difficulty in adapting boundaries to changing operating conditions, and training oscillations caused by inter-layer coupling and sample imbalance in constraints. To this end, this paper proposes an LLM-guided segmented reward reinforcement learning framework based on the HSRL framework [14]. By explicitly delineating different risk and profit zones through segment-based rewards, this approach utilizes a large language model (LLM) in conjunction with operational data to dynamically adjust segment boundaries, and employs a hierarchical dual-timescale objective network and a delayed strategy update to enhance training stability [15], [16].

A fixed penalty coefficient or a single continuous reward makes it difficult to simultaneously distinguish between the safe zone, the balanced zone, and the dangerous zone [17], which may result in the policy becoming overly conservative or frequently violating safety constraints. Segmented Reward Function Reinforcement Learning (PRRL) offers a highly promising solution to this problem. This method employs different reward function formulas based on control objectives in various environments; for example, it emphasizes utility in the safe zone, balances utility and constraints in the balanced zone, and guides rapid stabilization while adhering to fluctuation constraints in the hazardous zone. Experiments have demonstrated that in air combat decision-making [18] and mobile robot navigation and close-following scenarios [19], segmented reward models can significantly improve accuracy and stability in complex scenarios compared to single-value baseline reward models. On the other hand, Large Language

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Model-Enhanced Reinforcement Learning (LERL) provides a more efficient segmentation method for reinforcement learning agents using segmented reward functions. During training with segmented reward functions, precisely identifying the most appropriate segmentation boundaries is challenging due to the continuous nature of wind speeds in the WSIS environment. By using an LLM to fine-tune the segmentation boundaries based on data collected during training, an optimal partition can be obtained that balances constraint satisfaction with profit maximization. LLMs have been widely applied in various aspects of reinforcement learning [20], such as serving as information processors to reduce learning complexity [21], as reward designers to design reward functions in complex task environments [22], and as decision-makers to generate or guide reliable actions to accelerate offline reinforcement learning [23].

Although there has been extensive research in the academic community on both piecewise reward function reinforcement learning (PRRL) and LLM-enhanced RL (LERL), few studies have attempted to integrate PRRL and LERL to construct an LLM-enhanced piecewise reward function reinforcement learning (LEPRRL) framework. This integration presents several challenges, notably: 1) segmented reward functions introduce non-smooth reward surfaces at the boundaries, leading to high-variance gradient updates and cumulative overestimation; 2) fine-tuning of segmented boundaries by the LLM causes local shifts in the reward surface, resulting in Q-value estimates lagging behind surface changes and affecting the accuracy of policy updates.

To address these limitations, this paper proposes an LLM-enhanced segmented reward reinforcement learning (LEPRRL) framework based on the TD3 algorithm for the WSIS control problem, building upon the HSRL framework [14]. Its contributions are as follows:

1) A piecewise reward function method is proposed. By guiding the agent to pursue different objectives in different segments, this method improves control performance and operational safety, thereby resolving the challenge of balancing reward optimization with safety constraints.
2) A method for LLM-enhanced reinforcement learning is proposed. By utilizing an LLM to monitor parameters and fine-tune segmentation logic during training, this method identifies optimal segmentation points and optimizes the agent's overall performance.
3) To address the training instability introduced by the aforementioned segmented reward mechanism and LLM boundary fine-tuning, we introduce a TD3-based dual-Q-network and delayed policy update framework, which ensures the convergence stability of the LEPRRL framework during long-time-domain training.

II. The Design, Intent, and Limitations of the Templates

The templates are intended to **approximate the final look and page length of the articles/papers**. **They are NOT intended to be the final produced work that is displayed in print or on IEEEXplore®**. They will help to give the authors an approximation of the number of pages that will be in the final version. The structure of the LaTeX files, as designed, enable easy conversion to XML for the composition systems used by the IEEE. The XML files are used to produce the final print/IEEEXplore pdf and then converted to HTML for IEEEXplore.

III. Where to Get LaTeX Help — User Groups

The following online groups are helpful to beginning and experienced LaTeX users. A search through their archives can provide many answers to common questions.

http://www.latex-community.org/
https://tex.stackexchange.com/

IV. Other Resources

See [1]–[5] for resources on formatting math into text and additional help in working with LaTeX.

V. Text

For some of the remainer of this sample we will use dummy text to fill out paragraphs rather than use live text that may violate a copyright.

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$$x = \sum_{i=0}^{n} 2iQ. \tag{1}$$

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VI. Some Common Elements

A. Sections and Subsections

Enumeration of section headings is desirable, but not required. When numbered, please be consistent throughout the article, that is, all headings and all levels of section headings in the article should be enumerated. Primary headings are designated with Roman numerals, secondary with capital letters, tertiary with Arabic numbers; and quaternary with lowercase letters. Reference and Acknowledgment headings are unlike all other section headings in text. They are never enumerated. They are simply primary headings without labels, regardless of whether the other headings in the article are enumerated.

B. Citations to the Bibliography

The coding for the citations is made with the LaTeX `\cite` command. This will display as: see [1].

For multiple citations code as follows: `\cite{ref1,ref2,ref3}` which will produce [1]–[3]. For reference ranges that are not consecutive code as `\cite{ref1,ref2,ref3,ref9}` which will produce [1]–[3], [9]

C. Lists

In this section, we will consider three types of lists: simple unnumbered, numbered, and bulleted. There have been many options added to IEEEtran to enhance the creation of lists. If your lists are more complex than those shown below, please refer to the original "IEEEtran_HOWTO.pdf" for additional options.

A plain unnumbered list:

bare_jrnl.tex
bare_conf.tex
bare_jrnl_compsoc.tex
bare_conf_compsoc.tex
bare_jrnl_comsoc.tex



Fig. 1. Simulation results for the network.

A simple numbered list:

1) bare_jrnl.tex
2) bare_conf.tex
3) bare_jrnl_compsoc.tex
4) bare_conf_compsoc.tex
5) bare_jrnl_comsoc.tex

A simple bulleted list:

- bare_jrnl.tex
- bare_conf.tex
- bare_jrnl_compsoc.tex
- bare_conf_compsoc.tex
- bare_jrnl_comsoc.tex

D. Figures

Fig. 1 is an example of a floating figure using the graphicx package. Note that `\label` must occur AFTER (or within) `\caption`. For figures, `\caption` should occur after the `\includegraphics`.

Fig. 2(a) and 2(b) is an example of a double column floating figure using two subfigures. (The subfig.sty package must be loaded for this to work.) The subfigure `\label` commands are set within each subfloat command, and the `\label` for the overall figure must come after `\caption`. `\hfil` is used as a separator to get equal spacing. The combined width of all the parts of the figure should do not exceed the text width or a line break will occur.

Note that often IEEE papers with multi-part figures do not place the labels within the image itself (using the optional argument to `\subfloat[]`), but instead will reference/describe all of them (a), (b), etc., within the main caption. Be aware that for subfig.sty to generate the (a), (b), etc., subfigure labels, the optional argument to `\subfloat` must be present. If a subcaption is not desired, leave its contents blank, e.g.,`\subfloat[]`.

VII. Tables

Note that, for IEEE-style tables, the `\caption` command should come BEFORE the table. Table captions use title case. Articles (a, an, the), coordinating conjunctions (and, but, for,



(a) (b)

Fig. 2. Dae. Ad quatur autat ut porepel itemoles dolor autem fuga. Bus quia con nessunti as remo di quatus non perum que nimus. (a) Case I. (b) Case II.

TABLE I
AN EXAMPLE OF A TABLE

One	Two
Three	Four

or, nor), and most short prepositions are lowercase unless they are the first or last word. Table text will default to `\footnotesize` as the IEEE normally uses this smaller font for tables. The `\label` must come after `\caption` as always.

VIII. ALGORITHMS

Algorithms should be numbered and include a short title. They are set off from the text with rules above and below the title and after the last line.

Algorithm 1 Weighted Tanimoto ELM.

TRAIN($\mathbf{XT}$)
select randomly $W \subset \mathbf{X}$
$N_{\mathbf{t}} \leftarrow |\{i : \mathbf{t}_i = \mathbf{t}\}|$ **for** $\mathbf{t} = -1, +1$
$B_i \leftarrow \sqrt{\text{MAX}(N_{-1}, N_{+1})/N_{\mathbf{t}_i}}$ **for** $i = 1, ..., N$
$\hat{\mathbf{H}} \leftarrow B \cdot (\mathbf{X}^T\mathbf{W})/(\nVdash\mathbf{X} + \nVdash\mathbf{W} - \mathbf{X}^T\mathbf{W})$
$\beta \leftarrow \left(I/C + \hat{\mathbf{H}}^T\hat{\mathbf{H}}\right)^{-1} (\hat{\mathbf{H}}^T B \cdot \mathbf{T})$
return $\mathbf{W}, \beta$

PREDICT($\mathbf{X}$)
$\mathbf{H} \leftarrow (\mathbf{X}^T\mathbf{W})/(\nVdash\mathbf{X} + \nVdash\mathbf{W} - \mathbf{X}^T\mathbf{W})$
return SIGN($\mathbf{H}\beta$)

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IX. MATHEMATICAL TYPOGRAPHY AND WHY IT MATTERS

Typographical conventions for mathematical formulas have been developed to **provide uniformity and clarity of presentation across mathematical texts**. This enables the readers of those texts to both understand the author's ideas and to grasp new concepts quickly. While software such as LaTeX and MathType® can produce aesthetically pleasing math when used properly, it is also very easy to misuse the software, potentially resulting in incorrect math display.

IEEE aims to provide authors with the proper guidance on mathematical typesetting style and assist them in writing the best possible article. As such, IEEE has assembled a set of examples of good and bad mathematical typesetting [1]–[5].

Further examples can be found at http://journals.ieeeauthorcenter.ieee.org/wp-content/uploads/sites/7/IEEE-Math-Typesetting-Guide-for-LaTeX-Users.pdf

A. Display Equations

The simple display equation example shown below uses the "equation" environment. To number the equations, use the `\label` macro to create an identifier for the equation. LaTeX will automatically number the equation for you.

$$x = \sum_{i=0}^{n} 2iQ. \tag{2}$$

is coded as follows:

```
\begin{equation}
\label{deqn_ex1}
x = \sum_{i=0}^{n} 2{i} Q.
\end{equation}
```

To reference this equation in the text use the `\ref` macro. Please see (2)
is coded as follows:

```
Please see (\ref{deqn_ex1})
```

B. Equation Numbering

Consecutive Numbering: Equations within an article are numbered consecutively from the beginning of the article to the end, i.e., (1), (2), (3), (4), (5), etc. Do not use roman numerals or section numbers for equation numbering.

Appendix Equations: The continuation of consecutively numbered equations is best in the Appendix, but numbering as (A1), (A2), etc., is permissible.

Hyphens and Periods: Hyphens and periods should not be used in equation numbers, i.e., use (1a) rather than (1-a) and (2a) rather than (2.a) for subequations. This should be consistent throughout the article.

C. Multi-Line Equations and Alignment

Here we show several examples of multi-line equations and proper alignments.

A single equation that must break over multiple lines due to length with no specific alignment.

$$\begin{multline}\text{The first line of this example}\\ \text{The second line of this example}\\ \text{The third line of this example}\end{multline} \tag{3}$$

is coded as:

```
\begin{multline}
\text{The first line of this example}\\
\text{The second line of this example}\\
\text{The third line of this example}
\end{multline}
```

A single equation with multiple lines aligned at the = signs

$$\begin{aligned} a &= c+d \qquad (4)\\ b &= e+f \qquad (5)\end{aligned}$$

is coded as:

```
\begin{align}
a &= c+d \\
b &= e+f
\end{align}
```

The `align` environment can align on multiple points as shown in the following example:

$$\begin{aligned} x &= y & X &= Y & a &= bc \qquad (6)\\ x' &= y' & X' &= Y' & a' &= bz \qquad (7)\end{aligned}$$

is coded as:

```
\begin{align}
x &= y & X &=Y & a &=bc\\
x' &= y' & X' &=Y' &a' &=bz
\end{align}
```

D. Subnumbering

The amsmath package provides a `subequations` environment to facilitate subnumbering. An example:

$$\begin{aligned} f &= g \qquad (8a)\\ f' &= g' \qquad (8b)\\ \mathcal{L}f &= \mathcal{L}g \qquad (8c)\end{aligned}$$

is coded as:

```
\begin{subequations}\label{eq:2}
\begin{align}
f&=g \label{eq:2A}\\
f' &=g' \label{eq:2B}\\
\mathcal{L}f &= \mathcal{L}g \label{eq:2c}
\end{align}
\end{subequations}
```

E. Matrices

There are several useful matrix environments that can save you some keystrokes. See the example coding below and the output.

A simple matrix:

$$\begin{matrix} 0 & 1 \\ 1 & 0 \end{matrix} \tag{9}$$

is coded as:

```
\begin{equation}
\begin{matrix}  0 &  1 \\
1 &  0 \end{matrix}
\end{equation}
```

A matrix with parenthesis

$$\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \tag{10}$$

is coded as:

```
\begin{equation}
\begin{pmatrix} 0 & -i \\
 i &  0 \end{pmatrix}
\end{equation}
```

A matrix with square brackets

$$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \tag{11}$$

is coded as:

```
\begin{equation}
\begin{bmatrix} 0 & -1 \\
1 &  0 \end{bmatrix}
\end{equation}
```

A matrix with curly braces

$$\begin{Bmatrix} 1 & 0 \\ 0 & -1 \end{Bmatrix} \tag{12}$$

is coded as:

```
\begin{equation}
\begin{Bmatrix} 1 &  0 \\
0 & -1 \end{Bmatrix}
\end{equation}
```

A matrix with single verticals

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} \tag{13}$$

is coded as:

```
\begin{equation}
\begin{vmatrix} a &  b \\
c &  d \end{vmatrix}
\end{equation}
```

A matrix with double verticals

$$\begin{Vmatrix} i & 0 \\ 0 & -i \end{Vmatrix} \tag{14}$$

is coded as:

```
\begin{equation}
\begin{Vmatrix} i &  0 \\
0 & -i \end{Vmatrix}
\end{equation}
```

F. Arrays

The `array` environment allows you some options for matrix-like equations. You will have to manually key the fences, but there are other options for alignment of the columns and for setting horizontal and vertical rules. The argument to `array` controls alignment and placement of vertical rules.

A simple array

$$\left(\begin{array}{cccc} a+b+c & uv & x-y & 27 \\ a+b & u+v & z & 134 \end{array} \right) \tag{15}$$

is coded as:

```
\begin{equation}
\left(
\begin{array}{cccc}
a+b+c & uv & x-y & 27\\
a+b & u+v & z & 134
\end{array} \right)
\end{equation}
```

A slight variation on this to better align the numbers in the last column

$$\left(\begin{array}{cccc} a+b+c & uv & x-y & 27 \\ a+b & u+v & z & 134 \end{array} \right) \tag{16}$$

is coded as:

```
\begin{equation}
\left(
\begin{array}{cccr}
a+b+c & uv & x-y & 27\\
a+b & u+v & z & 134
\end{array} \right)
\end{equation}
```

An array with vertical and horizontal rules

$$\left(\begin{array}{c|c|c|r} a+b+c & uv & x-y & 27 \\ \hline a+b & u+v & z & 134 \end{array} \right) \tag{17}$$

is coded as:

```
\begin{equation}
\left(
\begin{array}{c|c|c|r}
a+b+c & uv & x-y & 27\\
a+b & u+v & z & 134
\end{array} \right)
\end{equation}
```

Note the argument now has the pipe "|" included to indicate the placement of the vertical rules.

G. Cases Structures

Many times cases can be miscoded using the wrong environment, i.e., `array`. Using the `cases` environment will save keystrokes (from not having to type the `\left\lbrace`) and automatically provide the correct column alignment.

$$z_m(t) = \begin{cases} 1, & \text{if } \beta_m(t) \\ 0, & \text{otherwise.} \end{cases}$$

is coded as follows:

```
\begin{equation*}
{z_m(t)} =
\begin{cases}
1,&{\text{if}}\ {\beta }_m(t),\\
{0,}&{\text{otherwise.}}
\end{cases}
\end{equation*}
```

Note that the "&" is used to mark the tabular alignment. This is important to get proper column alignment. Do not use `\quad` or other fixed spaces to try and align the columns. Also, note the use of the `\text` macro for text elements such as "if" and "otherwise."

H. Function Formatting in Equations

Often, there is an easy way to properly format most common functions. Use of the `\` in front of the function name will in most cases, provide the correct formatting. When this does not work, the following example provides a solution using the `\text` macro:

$$d_R^{KM} = \underset{d_l^{KM}}{\arg\min} \{ d_1^{KM}, \ldots, d_6^{KM} \}.$$

is coded as follows:

```
\begin{equation*}
 d_{R}^{KM} = \underset {d_{l}^{KM}}
 {\text{arg min}} \{ d_{1}^{KM},
 \ldots,d_{6}^{KM}\}.
\end{equation*}
```

I. Text Acronyms Inside Equations

This example shows where the acronym "MSE" is coded using \text{} to match how it appears in the text.

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2$$

```
\begin{equation*}
 \text{MSE} = \frac {1}{n}\sum _{i=1}^{n}
(Y_{i} - \hat {Y_{i}})^{2}
\end{equation*}
```

X. Conclusion

The conclusion goes here.

Acknowledgments

This should be a simple paragraph before the References to thank those individuals and institutions who have supported your work on this article.

Appendix
Proof of the Zonklar Equations

Use \appendix if you have a single appendix: Do not use \section anymore after \appendix, only \section*. If you have multiple appendixes use \appendices then use \section to start each appendix. You must declare a \section before using any \subsection or using \label (\appendices by itself starts a section numbered zero.)

References Section

You can use a bibliography generated by BibTeX as a .bbl file. BibTeX documentation can be easily obtained at: http://mirror.ctan.org/biblio/bibtex/contrib/doc/ The IEEEtran BibTeX style support page is: http://www.michaelshell.org/tex/ieeetran/bibtex/

Simple References

You can manually copy in the resultant .bbl file and set second argument of \begin to the number of references (used to reserve space for the reference number labels box).

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BIOGRAPHY SECTION

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